HARSHAD MADHAV DHUPPE

Software Developer | Python & Backend Engineer | Full-Stack Developer

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PROFESSIONAL SUMMARY

Final-year Electronics and Computer Engineering student who builds database-backed, API-driven applications with **Python, FastAPI, Flask, and SQL**. Delivered three end-to-end projects, two deployed live, featuring **REST APIs, authentication with role-based access control, AI and computer-vision integration, automated testing, and CI/CD**. Strong foundation in **Data Structures & Algorithms, OOP, and DBMS**. Seeking an entry-level **Software Engineer** / **Python Developer** role to build reliable, scalable backend systems.

EDUCATION

Bachelor of Engineering (B.E.) in Electronics and Computer Engineering — **Expected 2027**
CSMSS Chh. Shahu College of Engineering, Chhatrapati Sambhajinagar, Maharashtra — *CGPA: 7.46 / 10*

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, TypeScript, HTML5, CSS3
Backend & APIs: FastAPI, Flask, REST API Design, Pydantic, SQLAlchemy, Authentication & RBAC, Rate Limiting, Caching
Databases: SQLite, MySQL, PostgreSQL
Frontend: React, TypeScript, JavaScript, HTML5, CSS3
AI / ML & Computer Vision: OpenCV, NumPy, Pandas, Scikit-learn, Gemini API, Face Detection & Recognition (YuNet, SFace)
Code Quality & Security: Python AST, Bandit, Ruff, Input Validation, Error Handling, Debugging
DevOps & Testing: Git, GitHub, Docker, GitHub Actions (CI), Pytest, Playwright, API Testing
Core CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Software Testing

PROJECTS

CodeNiti AI – Automated Code Intelligence & Security Auditing Platform — GitHub | Live Demo

Tech Stack: *Python, FastAPI, AST, Bandit, Ruff, Gemini API, SQLite, HTML, CSS, JavaScript, Docker*

- Co-developed an AI-assisted code intelligence platform offering **code optimization, complexity analysis, debugging help, test generation, and security auditing**.
- Implemented deterministic **Python AST analysis** and **Bandit** security checks to detect vulnerabilities, code-quality issues, and maintainability risks.
- Built an automated **repair-verification workflow** that diffs code before and after AI-assisted fixes and re-scans the result with static analysis for remaining findings.
- Designed **FastAPI** endpoints with input validation, user authentication, private report history, request caching, rate limiting, and multi-file ZIP project auditing.
- Added automated tests and a **GitHub Actions CI pipeline** to validate API behavior and static-analysis components.

ClassLens – Vision-Based Smart Attendance & Classroom Monitoring System — GitHub | Live Demo

Tech Stack: *Python, FastAPI, OpenCV, React, TypeScript, SQLite, HTML, CSS, Docker*

- Developed a full-stack attendance system that uses **computer vision** to detect and recognize registered students from classroom images and camera captures.
- Integrated **OpenCV YuNet and SFace** models with similarity-based matching and a teacher-review step for low-confidence identifications.
- Implemented subject-wise sessions, duplicate-attendance prevention, persistent student records, and **role-based dashboards** for administrators, teachers, and students.
- Built attendance-correction and leave-request workflows, student attendance insights, and **CSV report generation**.
- Added API tests, **browser-based end-to-end tests**, input validation, and **Docker** setup for reliable, reproducible deployment.

KrishiAI – AI-Based Organic Farming Advisory System & Marketplace — GitHub

Tech Stack: *Python, Flask, SQLAlchemy, Scikit-learn, Gemini API, SQLite / PostgreSQL, HTML, CSS, JavaScript*

- Built a web platform combining **crop recommendation, plant-disease advisory, weather data, and an online marketplace** for farmers and buyers.
- Integrated a **Random Forest** crop-recommendation model using soil nutrients, temperature, humidity, pH, and rainfall as input features.
- Implemented image-based disease advisory via **Gemini API** with suggested treatments and agricultural guidance.
- Developed role-based farmer/buyer accounts, product management, shopping cart, and order workflows with **Flask and SQLAlchemy**.
- Integrated an AI chatbot, multilingual interface, and analytics dashboards to improve accessibility and usability.

CERTIFICATIONS

Python Programming – GeeksforGeeks | **C Programming** – NPTEL | **SQL** – Great Learning